# 27. Central configurations in a bounded, symmetry-reduced N-body problem

Pure Thought research portfolio

Revised 17 September 2026

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**Revised specification — 2026-09-11.** Detailed critique · Original description · Evidence standard

## Plain-language guide

### The problem in everyday terms

Most collections of gravitating bodies change their overall shape as they move. Certain special arrangements can rotate or expand and contract while keeping the same shape. These are central configurations. This problem asks for every such arrangement in a carefully limited case, with a proof that the list is complete and a separate analysis of what small disturbances do to the associated rotating motion.

### Key terms

- **N-body problem** — The problem of describing the motion of N mutually gravitating bodies.
- **Central configuration** — An arrangement in which each body's gravitational acceleration is proportional to its position relative to the center of mass, with one common proportionality factor.
- **Homographic motion** — Motion preserving the arrangement's shape while its size and orientation may change.
- **Relative equilibrium** — A motion that is stationary in an appropriate rotating reference frame.
- **Center of mass** — The mass-weighted average position, used here to remove overall translation.
- **Normalization** — A fixed choice of scale, such as setting the moment of inertia to one, so scaled copies are not counted separately.
- **Symmetry quotient** — The convention for treating configurations related by rotations, reflections or specified relabelings as equivalent.
- **Collision-free** — Having strictly positive separation between every pair of bodies.
- **Coriolis term** — A velocity-dependent term arising in a rotating frame, essential to the stability calculation.
- **Spectral stability** — A statement about eigenvalues of the linearized dynamics; it is weaker than a full nonlinear stability proof.

### Why this matters

Central configurations are organizing structures in gravitational dynamics and provide exact reference cases amid otherwise complicated motion. A complete small catalog can test numerical methods and reveal how mass choices and symmetry affect possible motions. The problem also links gravity to exact algebraic root finding.

### What progress would mean

Progress would establish a complete list under a stated mass choice and symmetry convention. It would not classify every periodic orbit: a choreography such as the figure-eight changes shape and belongs to a different question.

## Core question

For fixed positive masses and a declared symmetry quotient, can all collision-free central configurations in the chosen family be certified and their relative-equilibrium linearization analyzed?

## Scope and assumptions

Start with three labeled positive rational masses in the plane. Fix G=1, center of mass zero, moment of inertia I=Σm_i|r_i|$^2$=1, and quotient by O(2), so mirror images count together. Permute only equal masses when explicitly requested. Extend to a specified four-body symmetry family after validating the complete three-body case.

Central configurations generate homographic motions; general periodic choreographies are a separate problem.

## Mathematical target

Use positive U=Σ_{i<j}m_im_j/|r_i−r_j| and

$$\nabla_{r_i} U = -\lambda m_i r_i, \qquad \lambda = U/I > 0.$$

Introduce distance variables d_ij>0 with d_ij$^2$=|r_i−r_j|$^2$, and retain positivity/collision exclusions when polynomializing. Use symmetry charts covering every configuration claimed. Completeness requires certified global domain coverage or an exact elimination/root-count argument, not just isolated numerical solutions.

## Required outputs and proof obligations

Return normalized configurations, distance enclosures, residual identities, existence/uniqueness proofs, equivalence representatives, and completeness scope. For planar relative equilibria analyze the reduced rotating-frame first-order system including Coriolis terms. Distinguish spectral stability from nonlinear stability.

## Validation and rejection controls

For three labeled masses under the stated O(2) quotient, recover three collinear orderings (one per middle body) and one equilateral shape. Check λ=U at I=1. Test collision artifacts introduced by denominator clearing. For stability, reproduce a known three-body criterion using identical mass and reduction conventions.

## Milestones and research extension

Baseline: complete certified three-body catalog. Strong: complete enumeration in one four-body symmetry family or mass instance, with all charts covered. Research extension: a five-body re-

stricted family; no claim to solve the general finiteness problem without an appropriately global proof.

Milestones are acceptance gates, not calendar promises. Before a run, freeze the concrete instance/domain, tolerance, compute budget, and permitted dependencies. A bounded search that fails to find a result must preserve an unresolved outcome.

### Starting instruction

Work on the baseline above. First make every input and convention explicit, then validate the reference case. Build the checker alongside the calculation. Label numerical evidence, finite-model certificates, and continuum theorems separately; return the strongest justified result and identify the exact unmet proof obligation.

### Primary references and their role

- Yu and Zhu, On the finiteness of four-body central configurations — Context for finiteness versus explicit certified enumeration.

References support definitions or limitations; the scope choices and proposed milestones are editorial recommendations, not claims that these results have already been obtained.

### Displayed formulation and claim boundary

Canonical specification.

$$U = \sum_{i<j} \frac{m_i m_j}{|r_i - r_j|}, \quad \sum_i m_i r_i = 0, \quad I = \sum_i m_i |r_i|^2 = 1,$$

$$\nabla_{r_i} U = -\lambda m_i r_i, \qquad \lambda = U/I > 0, \qquad d_{ij}^2 = |r_i - r_j|^2, \quad d_{ij} > 0.$$

Preserve collision exclusions when clearing denominators. For three labeled positive masses modulo $O(2)$, recover three collinear orderings and one equilateral shape. Complete coverage requires more than isolated root boxes. Relative-equilibrium stability uses the reduced rotating-frame system including Coriolis terms; a figure-eight choreography is not a central configuration.

### Detailed review and rationale

Revised problem · Preserved original · Pinned upstream source

Line references below refer to the preserved original at commit `f854d4e12b55`, not the rewritten file. "Blocking" means the original issue can invalidate the scientific target or claimed certificate; "major" means it materially changes scope, inference, or acceptance criteria. These are critiques of specifications, not results of running the embedded research-code sketches.

#### Detailed findings

1. **Blocking — wrong example (26–29, 120).** The figure-eight choreography changes shape and is not a central configuration. It belongs in a periodic-orbit problem, retained separately in the supplement.

2. **Blocking — counts and quotients (112–114).** The text confuses five restricted-problem Lagrange points with Euler central configurations. Counting depends on labels, rotations and reflections; the rewrite states the quotient and a four-shape three-body benchmark.
3. **Blocking — stability shortcut (124–130).** Hessian signs alone do not give rotating-frame orbital stability; Coriolis terms and symmetry modes must be treated.
4. **Major — algebraic and finiteness claims (36, 93–98).** Clearing denominators can add collisions and does not remove square roots without auxiliary variables. Numerical roots do not prove completeness, and the generic-five-body status claim needs qualification against the literature. The new bounded scope avoids an unsupported universal claim.

#### What the revision preserves and changes

The central motivation is retained, but the executable question is now: **For fixed positive masses and a declared symmetry quotient, can all collision-free central configurations in the chosen family be certified and their relative-equilibrium linearization analyzed?** The revised scope chooses a concrete starting model; those choices are proposed research-design decisions, not facts inferred from the original. The baseline can succeed without resolving the full open research program.

The replacement also removes the original implementation sketches from the active specification. They remain in the archive for comparison, but have not been made executable or certified. New implementations should derive their equations from the revised definitions rather than copy unchecked prototypes.

#### Acceptance criteria to use instead

Baseline: complete certified three-body catalog. Strong: complete enumeration in one four-body symmetry family or mass instance, with all charts covered. Research extension: a five-body restricted family; no claim to solve the general finiteness problem without an appropriately global proof.

#### Sources supporting the corrections

- Yu and Zhu, On the finiteness of four-body central configurations — Context for finiteness versus explicit certified enumeration.

## Evidence and execution standard

This standard applies to the revised specifications. It is a proposed contract for future research runs; this editorial revision does not supply the scientific certificates themselves.

### What "pure thought" means here

Use explicitly stated mathematical models and deductions without fitting conclusions to experimental outcomes. Synthetic examples, exact model inputs, analytic references, numerical search, and established software are allowed. Physical constants, electronic-structure approximations, constitutive laws, and empirical parameter sets must be identified when used. A model-derived result is conditional on that model.

The original ban on legacy software conflicts with its proposed reliance on NumPy, SciPy, CVXPY, nauty, and other libraries. Reimplementing every optimizer and algebra system is not necessary for scientific independence. Prefer a small independently checkable certificate and a documented trusted base. A fresh implementation can be a project objective, but must not be confused with a mathematical proof or novelty claim.

### Four evidence levels

| Label | What it establishes | What it does not establish |
|---|---|---|
| Numerical evidence | A reproducible computation at stated precision, sample size, and discretization | Exact truth, continuum coverage, or a universal theorem |
| Finite-model certificate | An exact/validated statement about the specified finite model or finite relaxation | An omitted tail, thermodynamic limit, or physical realization |
| Conditional mathematical theorem | A proof over the stated domain with every hypothesis and error controlled | Truth outside those assumptions or a demonstrated physical system |
| Formal proof | A checked theorem in a named proof assistant, with imported axioms/dependencies listed | Correct physical modeling merely by virtue of formalization |

Formalization is an optional extra deliverable unless the specific project makes it mandatory. High precision, JSON output, solver status and small residuals do not by themselves confer any of these proof levels.

### Claim directions

For minimization over a larger feasible set, the optimum is a lower bound on the original minimum. A physical feasible point supplies an upper bound. For maximization the directions reverse. Track feasibility of the primal/dual witnesses separately from their objective values.

Infeasibility of an inner ansatz excludes only that ansatz. Infeasibility of an outer relaxation can exclude the original problem if the map into the relaxation is proved and all approximations preserve necessity. Passing necessary tests means “not excluded”; existence needs a constructive sufficient argument.

A smallest observed example is not a minimality theorem. A complete catalog must define the universe, equivalence relation, boundaries, and proof that pruning misses nothing. A finite-size numerical trend does not prove an asymptotic limit.

### Minimum run manifest

Before calculation, record the problem ID and revision hash, exact inputs and conventions, quantified domain, equivalence relation, target observable, accepted evidence level, tolerances, and resource limits. Record dependencies and precision/rounding mode. Where randomization is used, preserve seeds, trial counts and raw sufficient statistics. Resource limits are planning decisions, not evidence that a mathematical solution cannot exist.

An output bundle should include:

- The statement being claimed, including all assumptions and quantifiers.
- Construction or proof object, plus an independently runnable verifier and its trust assumptions.
- Numerical/error ledger separating roundoff, discretization, truncation, optimization, sampling and modeling error where relevant.
- Positive reference cases and negative controls that the verifier must reject.
- Coverage: included domain, omitted domain, unresolved regions and termination reason.

These are requirements for future implementations, not a promise of polynomial-size certificates for every problem.

### Checking specific certificate types

**LP/SDP/SOS:** verify equality constraints, objective direction, and cone feasibility using exact arithmetic or outward-rounded intervals. Rationalizing a numerical solution can break PSD constraints. Correct residuals with justified norm bounds; do not silently project and reuse the old objective. Distinguish a proof of a polynomial approximation from a proof for the original function.

**Interval numerics:** state function enclosures, domains and outward rounding. A small root box needs existence and, if claimed, uniqueness. Cover the complement for a global enumeration. Include singularities and tails rather than discarding them numerically.

**SAT/SMT:** a solver's UNSAT label is not a certificate. Preserve a proof supported by a checker and the proof that the encoding, symmetry reduction and blocked solutions match the mathematical universe. DRAT applies to an appropriate Boolean CNF proof, not automatically to every real-arithmetic or SDP claim.

**Statistics:** specify the estimand, dependence assumptions, confidence procedure and bias. Zero observed failures does not prove zero failure probability. MCMC error bars based on estimated autocorrelation are not generally rigorous mixing guarantees.

### Milestones and novelty

Baseline means a working, independently validated result on a concrete reference problem. Strong means a broader or sharper result with its claimed evidence level. Research extension means a possible novel result whose feasibility and significance need review. A negative theorem can be valuable; an unsuccessful search can also be useful if its scope is documented, but the two are not interchangeable.

Do not promise "publication-quality" in a fixed number of months. Before claiming novelty, conduct a focused current literature comparison against the exact theorem/model/metric. The sources in this revision check key definitions and limitations; they are not an exhaustive novelty audit across all 30 fields.